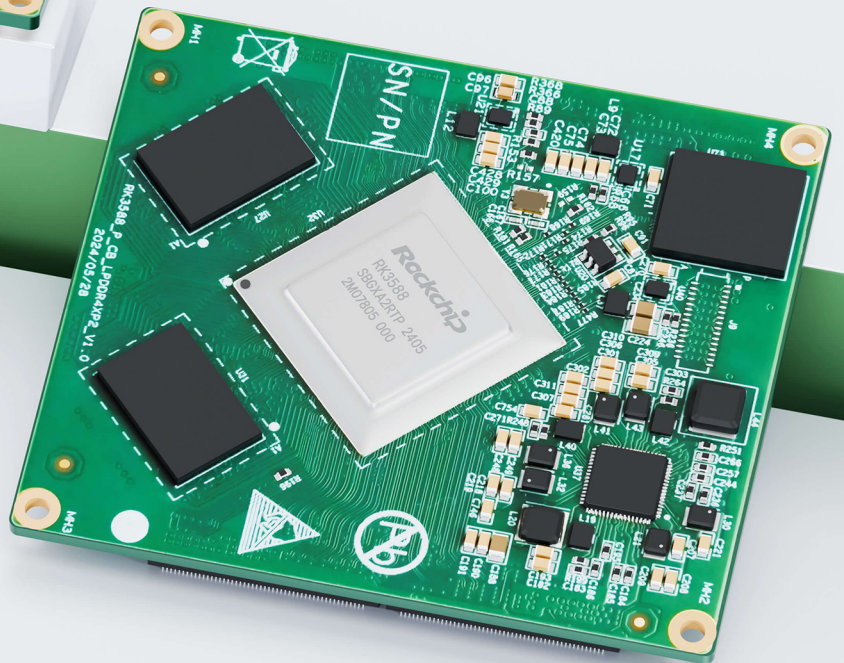
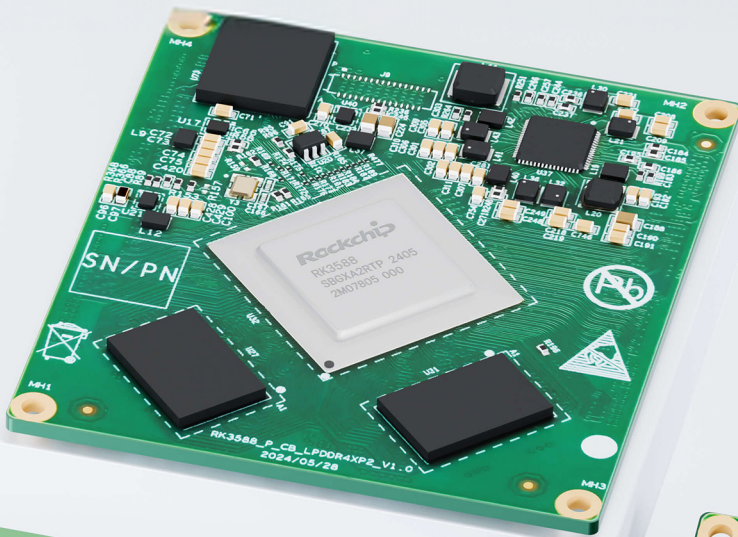


# Embrace Edge AI, Empower Industrial Digitalization



## EC3588-C AI System on Module

· RK3588 ARM SoC

· 4×A76+4×A55

· 6TOPS NPU

· 8K Video Codec

# 1. Product Overview

EC3588-C system on module is developed based on the Rockchip RK3588 flagship ARM SoC, delivering powerful performance for high-end embedded applications.

Product Features:

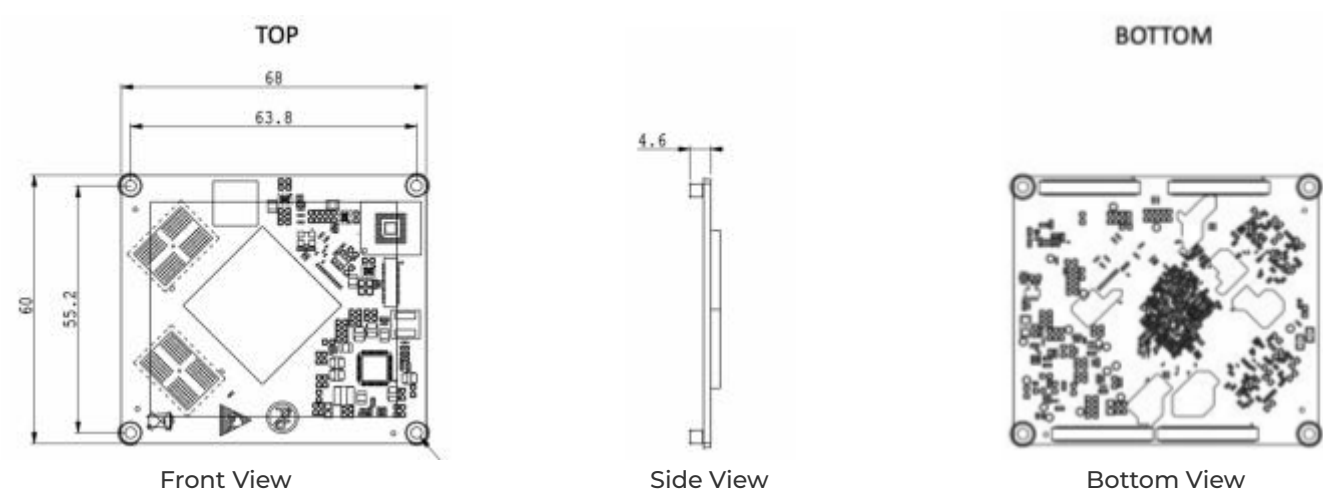
- **High-Performance CPU:** 8nm octa-core, 4×A76+4×A55 @ 2.4 GHz
- **Powerful AI NPU:** 6.0 TOPS triple-core, multi-format acceleration
- **Advanced Graphics:** Mali-G610 MP4, 8K@60fps codec, quad-screen output
- **Rich Interfaces:** PCIe 3.0, USB 3.1, SATA, GMAC, MIPI-CSI/DSI
- **High-End Media:** HDMI RX/TX, DP TX, multi-camera support

## Core Technical Specification

| Technical Indicator | Specification                                                     |
|---------------------|-------------------------------------------------------------------|
| OS                  | Linux                                                             |
| Industrial Protocol | Modbus RTU/TCP, EtherNet/IP, OPC UA, Mitsubishi MC                |
| Remote Management   | InHand DeviceLive, HTTPS, SSH                                     |
| AI                  | 6.0 TOPS NPU                                                      |
| Video Codec         | 8K@60 fps decode, 8K@30 fps encode                                |
| Cloud Platform      | Cloud parameter config, container management, firmware management |
| CPU                 | 4×A76 @ 2.4 GHz + 4×A55 @ 1.8 GHz                                 |
| Memory / Storage    | 16 GB LPDDR4 / 64 GB eMMC                                         |
| Display Interface   | HDMI/eDP TX, DP TX, MIPI DSI, quad independent display            |

| Technical Indicator    | Specification                             |
|------------------------|-------------------------------------------|
| Connectivity           | 2×GMAC, USB 3.1, PCIe 3.0, SATA, MIPI-CSI |
| Dimensions (W × D × H) | 68 × 60 × 4.6 mm                          |
| Operating Temperature  | 0 °C ~ +80 °C                             |

## 2. Product Dimensions



- Note:
- 1. All dimensions are in millimeters (mm).
  - 2. All dimensions are approximate and for reference only.
  - 3. Dimensioned drawings are not intended for machining.
  - 4. Dimensions are subject to part and manufacturing tolerances.
  - 5. Specifications may change without prior notice.

## 3. Hardware Specifications

| Category/Parameter | Specification |
|--------------------|---------------|
| Hardware Platform  |               |

| Category/Parameter  | Specification                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| CPU                 | 4 × ARM Cortex-A76 @ 2.4 GHz + 4 × ARM Cortex-A55 @ 1.8 GHz                                                                               |
| NPU                 | 6.0 TOPS, triple core, supports int4/int8/int16/FP16/BF16/TF32 acceleration                                                               |
| GPU                 | ARM Mali-G610 MP4, OpenGL ES 1.1/2.0/3.2, OpenCL 2.2, Vulkan 1.2                                                                          |
| VPU                 | Decode: H.265/VP9 up to 8K@60 fps, H.264 up to 8K@30 fps, AV1 up to 4K@60 fps; Encode: H.265/HEVC, H.264/AVC up to 8K@30 fps              |
| RAM                 | 16 GB LPDDR4                                                                                                                              |
| ROM                 | 64 GB eMMC                                                                                                                                |
| <b>Video Input</b>  |                                                                                                                                           |
| MIPI DCPHY          | 2; Supports DPHY or CPHY; 4-lane MIPI DPHY V2.0 (4.5 Gbps/lane); 3-lane MIPI CPHY V1.1 (2.5 Gbps/lane); Supports multiple MIPI-CSI combos |
| MIPI CSI DPHY       | 4; 2-lane MIPI DPHY V1.2 (2.5 Gbps/lane); Two 2-lane can be combined to 4-lane                                                            |
| DVP                 | 1; 8/10/12/16-bit standard DVP up to 150 MHz; Supports BT.601/BT.656/BT.1120                                                              |
| HDMI RX             | 1; Supports HDMI 2.0 (3.4 ~ 6 Gbps) and HDMI 1.4b (250Mbps ~ 3.4 Gbps); HDCP2.3 and HDCP1.4                                               |
| <b>Video Output</b> |                                                                                                                                           |
| HDMI/eDP TX         | ≤2; HDMI up to 7680 × 4320@60 Hz (3/6/8/10/12 Gbps), HDCP2.3; eDP up to 4K@60 Hz (1.62/2.7/5.4 Gbps), HDCP1.3                             |

| Category/Parameter  | Specification                                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| DP TX               | 2; DP TX 1.4a, available for USB3.1 Gen 1, supports 1/2/4 lanes; Up to 8192 × 4320@30 Hz; HDCP2.3, HDCP1.3                             |
| MIPI DSI            | 2; MIPI DPHY 2.0 or CPHY 1.1, up to 4K@60 Hz; Dual MIPI-DSI left/right mode, RGB/YUV up to 10-bit                                      |
| BT.1120             | 1; RGB up to 8-bit up to 150 MHz; Up to 1920 × 1080@60 Hz                                                                              |
| <b>Audio</b>        |                                                                                                                                        |
| I2S                 | ≤4; I2S0/I2S1: 8 lanes, TX/RX, 16-32 bits, up to 192 kHz; I2S2/I2S3: 2 lanes, TX/RX, 16~32 bits, up to 192 kHz                         |
| SPDIF               | 2; 2× 16-bit data storing; Biphasic stereo output                                                                                      |
| PDM                 | 2; Up to 8 channels, 16~24 bits, up to 192 kHz; PDM primary receive mode                                                               |
| DSM PWM             | 1; Convert PCM to bitstream digital 1-bit output                                                                                       |
| <b>Connectivity</b> |                                                                                                                                        |
| Ethernet            | 2 × GMAC by RGMII/RMII, 10/100/1000 Mbps                                                                                               |
| USB 3.1 Gen 1       | 3; Up to 5 Gbps; 2 × USB3.1 OTG (multiplexed with DP TX, supports USB Type-C and DP Alt); 1 × USB3.1 Host (multiplexed with PIPE PHY2) |
| USB 2.0 Host        | 2                                                                                                                                      |
| PCIe 3.0            | 2; Supports RC and EP, up to 8 Gbps; Combinations: 1×4, 2×2, 4×1, 1×2+2×1                                                              |
| PCIe 2.0            | ≤3; Each supports 1 lane, up to 5 Gbps                                                                                                 |
| SATA                | ≤3; 3 × SATA3.0 controllers, multiplexed by PCIe and USB_HOST2; Supports eSATA up to 6 Gbps                                            |

| Category/Parameter     | Specification                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------|
| UART                   | ≤10; 2 × 64-bit FIFO for TX/RX; 5/6/7/8-bit serial, up to 4 Mbps; All 10 UARTs support auto flow control |
| SPI                    | ≤5; Each controller supports two chip select outputs; Master and slave mode configurable                 |
| I2C                    | ≤9; 7-bit and 10-bit address modes; Standard mode up to 100 kbps, high-speed mode up to 400 kbps         |
| PWM                    | ≤16; Supports capturing mode                                                                             |
| ADC                    | ≤8; 8 × 12-bit single-end input SAR-ADC, up to 1 MS/s                                                    |
| <b>Power</b>           |                                                                                                          |
| Input Power            | DC 4V                                                                                                    |
| <b>Mechanical</b>      |                                                                                                          |
| Dimensions (W × D × H) | 68 × 60 × 4.6 mm                                                                                         |
| Package                | Board-to-board connector (4 × 100-pin, 0.4 mm pitch, combined height 1.5 mm)                             |
| Mounting Holes         | 4 × φ2.5 mm                                                                                              |
| <b>Environment</b>     |                                                                                                          |
| Operating Temperature  | 0 °C ~ +80 °C                                                                                            |

## 4. Software Specifications

| Category/Parameter      | Specification |
|-------------------------|---------------|
| <b>Operating System</b> |               |



| Category/Parameter                     | Specification                                                                                                                                                                  |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OS                                     | Linux                                                                                                                                                                          |
| OS Flashing Method                     | USB OTG                                                                                                                                                                        |
| <b>Data Acquisition Protocol (DSA)</b> |                                                                                                                                                                                |
| Industrial Protocol                    | Modbus RTU Master/Slave, Modbus TCP Master/Slave, EtherNet/IP, ISO on TCP, OPC UA Client/Server, Mitsubishi MC 3C/3E/3C OverTCP, Mitsubishi CPU Port, FINS UDP, Host Link, PPI |
| Electricity Protocol                   | DLT645-2007, IEC101/104, DNP3.0                                                                                                                                                |
| Other Protocol                         | BACnet, CNC                                                                                                                                                                    |
| <b>Maintenance and Management</b>      |                                                                                                                                                                                |
| Upgrade Method                         | Supports patent upgrade mechanism, local or remote firmware upgrade                                                                                                            |
| Log                                    | Supports local system logs, remote logs, important log power-off preservation                                                                                                  |
| Remote Management                      | InHand DeviceLive, HTTP, HTTPS, SSH, etc.                                                                                                                                      |
| DeviceLive Cloud                       | Supports cloud-based parameter configuration, container management, application and firmware management                                                                        |

## 5. Ordering Information

### Model Code

Model code: EC3588-C

## Product Model

| Model        | CPU                        | CPU Clock Speed                 | RAM      | ROM      | Operating Temperature |
|--------------|----------------------------|---------------------------------|----------|----------|-----------------------|
| EC358<br>8-C | 4 ×<br>A76 +<br>4 ×<br>A55 | A76 @ 2.4 GHz,<br>A55 @ 1.8 GHz | 16<br>GB | 64<br>GB | 0 °C ~ +80 °C         |

## 6. Contact Us

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